TRANSPORT MODULE (FLEET & STUDENT LOGISTICS ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Transport Managers • Drivers • IT Team • Parents • Students
Related Modules: Student Management • Attendance • Notifications • Parent Portal • Analytics • Finance

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1. MODULE OVERVIEW

1.1 Purpose

The Transport Module manages all school transport operations, including fleet management, route planning, student allocation, and real-time tracking. It enables schools to:

- Manage vehicles, drivers, and routes
- Assign students to transport services
- Track buses in real time
- Monitor attendance during transport
- Handle incidents and maintenance
- Provide transparency and safety for parents

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2. CORE OBJECTIVES

1. Provide centralized fleet and route management
2. Enable student assignment to buses and stops
3. Deliver real-time vehicle tracking and ETA updates
4. Ensure student safety through monitoring and alerts
5. Automate maintenance and incident management workflows
6. Provide analytics for transport efficiency and performance
7. Support mobile-first parent and driver interaction
8. Ensure compliance with safety and transport regulations

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Vehicle Registered → Route Created → Stops Defined → Students Assigned → Transport Active → Trips Monitored → Incidents Logged → Maintenance Scheduled → Reports Generated

3.2 Daily Operational Flow

Driver starts route
↓
Students board bus (attendance recorded)
↓
System tracks vehicle location in real time
↓
Parents receive ETA and notifications
↓
Incidents logged (if any)
↓
Trip completed and attendance finalized
↓
Analytics and logs updated

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4. MODULE ARCHITECTURE

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Transport Module
│
├── Fleet Management
├── Route & Stop Management
├── Student Assignment
├── Real-Time Tracking (GPS)
├── Attendance Tracking (Transport)
├── Incident Management
├── Maintenance Management
├── Parent Portal Integration
├── Notifications & Alerts
├── Analytics & Reporting
├── Mobile & Driver Interface
├── Security & Compliance
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5. FLEET MANAGEMENT

5.1 Features

- Register and manage vehicles
- Track vehicle status (active, maintenance, inactive)
- Record seating capacity and utilization
- Assign drivers to vehicles

5.2 Data Model

Vehicle

- vehicle_id
- registration_number
- capacity
- status (active/maintenance/inactive)
- driver_id

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6. ROUTE & STOP MANAGEMENT

6.1 Features

- Define routes with multiple stops
- Map routes using GPS coordinates
- Estimate travel time and distances
- Assign routes to vehicles

6.2 Data Model

Route

- route_id
- name
- stops (array)
- assigned_vehicle_id
- estimated_duration

Stop

- stop_id

- route_id
- location
- sequence_order

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7. STUDENT ASSIGNMENT

7.1 Features

- Assign students to routes, stops, and vehicles
- Track enrollment in transport services
- Link with Student Management Module

7.2 Data Model

TransportAssignment

- assignment_id
- student_id
- route_id
- stop_id
- vehicle_id
- status

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8. REAL-TIME TRACKING (GPS)

8.1 Features

- Live vehicle tracking on map
- ETA calculation for each stop
- Parent visibility via portal
- Geo-fencing for route monitoring

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9. TRANSPORT ATTENDANCE TRACKING

9.1 Features

- Record boarding and drop-off attendance
- QR code or RFID-based check-in

- Link attendance to Student Module

9.2 Rules

IF student misses pickup
→ Notify parent
→ Log absence

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10. INCIDENT MANAGEMENT

10.1 Features

- Log incidents (accidents, delays, misconduct, breakdowns)
- Categorize by severity (low, medium, high)
- Track resolution status and timelines
- Notify relevant stakeholders

10.2 Data Model

Incident

- incident_id
- vehicle_id
- type
- severity
- description
- status
- reported_by
- resolution_notes

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11. MAINTENANCE MANAGEMENT

11.1 Features

- Schedule preventive maintenance
- Track service history per vehicle
- Automated reminders for inspections
- Log repairs and downtime

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12. PARENT PORTAL INTEGRATION

12.1 Features

- Live tracking of assigned bus
- ETA notifications
- Home location pin setup
- Alerts for delays or incidents
- Emergency contact integration

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13. NOTIFICATIONS & ALERTS

13.1 Features

- Real-time alerts for route start/end
- Delay notifications
- Incident alerts
- Attendance notifications (pickup/drop-off)

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14. ANALYTICS & REPORTING

14.1 Features

- Route efficiency and travel time analysis
- Vehicle utilization (seat occupancy rates)
- Driver performance metrics
- Incident frequency and resolution tracking

14.2 Dashboards

- Fleet performance overview
- Student transport usage
- Safety and incident reports

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15. MOBILE & DRIVER INTERFACE

15.1 Features

- Driver mobile app for route navigation
- Attendance marking via mobile device
- Real-time updates and alerts
- Offline mode with sync

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16. SECURITY & COMPLIANCE

16.1 Features

- Role-based access control
- Driver verification and license tracking
- Secure GPS data handling
- Emergency protocols and SOS alerts

16.2 Compliance

- Adherence to local transport and child safety regulations
- Data protection for student information

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17. SCALABILITY

- Support large fleets and multiple routes
- Multi-campus and multi-tenant support
- High-performance real-time tracking

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18. IMPLEMENTATION GAPS (CRITICAL FIXES)

18.1 Current Issues Identified

- No students assigned to transport system
- Limited incident management workflow
- Missing maintenance logs and scheduling
- Redundant UI elements
- Lack of analytics and reporting
- Weak visibility of safety and compliance features

18.2 Required Fixes

- Implement student assignment workflows
- Expand incident management with severity and resolution tracking
- Add maintenance scheduling and logs
- Improve UI/UX (remove redundancy, add indicators)
- Build analytics dashboards
- Integrate safety and compliance tracking

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19. FUTURE ENHANCEMENTS

- AI-based route optimization
- Fuel tracking and cost optimization
- Smart attendance via facial recognition
- Integration with city traffic systems
- Advanced predictive maintenance